

CALORIE & NUTRITION TRACKER



Features Required:

1. Log daily food intake with meal categories (Breakfast, Lunch, Dinner, Snacks)
2. Display nutritional information: Calories, Protein, Carbs, Fats per food item
3. Calculate and display total daily nutrition summary
4. Track nutrition across multiple days using file storage
5. Provide daily calorie goal comparison and progress display
6. Allow searching food items from a predefined nutrition database

Concepts to be Used: Arrays, Structures, Loops, Conditional Statements, File Handling, String Functions, Standard Input / Output Functions

Technology Used: C Programming Language

Output:

Console-based Calorie & Nutrition Tracker

Daily nutrition summary displayed after logging meals

CALORIE & NUTRITION TRACKER

A C Programming Mini Project

1. Introduction

The Calorie & Nutrition Tracker is a console-based application developed using the C programming language. The main objective of this project is to help users monitor their daily food intake and nutritional consumption across categories such as Calories, Protein, Carbohydrates, and Fats. The program is menu-driven, allowing the user to log meals, view nutritional summaries, and compare intake against daily goals. Each food item is stored with its nutritional values. The user logs meals by selecting food items, and the system calculates and displays the cumulative nutrition. This project demonstrates the use of structures, file handling, loops, and standard input/output functions in C programming.

2. Objectives

- To develop a menu-driven nutrition tracking application using C
- To maintain a predefined food database with nutritional values
- To implement meal logging across Breakfast, Lunch, Dinner, and Snacks
- To calculate total daily calorie and macro-nutrient intake
- To compare daily intake against a user-defined calorie goal
- To store and retrieve food logs using file handling
- To improve logical thinking and programming skills

3. System Features

3.1 Food Database

The system maintains a built-in database of common food items with their calorie, protein, carbohydrate, and fat values per serving. The user searches and selects items from this database when logging meals.

3.2 Meal Logging

The program displays a menu with four meal categories: Breakfast, Lunch, Dinner, and Snacks. The user selects a category, then chooses food items to add. Multiple items can be added per meal. The meal data is saved to a file for persistence.

3.3 Nutrition Calculation

The user's input is accumulated across all meals. Calories, protein, carbs, and fats are summed and displayed as a daily total. If the total exceeds the user's daily calorie goal, a warning is shown.

3.4 Daily Summary & Goal Tracking

After logging meals, the system displays a complete nutritional breakdown. The summary shows total intake, remaining calories, and percentage of goal achieved.

4. Technology & Concepts Used

4.1 C Programming Concepts

- Structures (struct) – to store food item data (name, calories, protein, carbs, fats)
- Switch-case – to select meal categories and menu options
- If-else statements – to validate input and compare against goal
- Arrays – to store food database and logged meals
- File Handling – to save and load daily logs (fopen, fprintf, fscanf)
- Standard I/O functions – printf() and scanf()

4.2 Program Control Technique

The program uses a menu-driven approach where the user's selection controls the execution flow. Each meal category is implemented as a separate case inside the switch statement. A do-while loop keeps the main menu active until the user chooses to exit.

5. System Design

5.1 Data Used

- struct Food { char name[50]; int calories; float protein, carbs, fats; } – food item structure
- int choice – stores selected menu option
- int totalCalories – accumulates daily calorie count
- float totalProtein, totalCarbs, totalFats – accumulate macronutrient totals
- int goal – stores user-defined daily calorie goal
- FILE *fp – file pointer for log storage

These variables help in controlling the tracker flow and nutrition calculation.

6. Workflow

Main Program Flow:

1. Display main menu (Log Meal / View Summary / Set Goal / Exit)
2. Accept user category choice
3. Display food database for selected meal category
4. Accept user food selections and quantities
5. Calculate nutrition values for selected items
6. Accumulate totals and save to log file
7. Display daily nutrition summary with goal comparison

7. Sample Output (Console)

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==== CALORIE & NUTRITION TRACKER ====
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1. Log Meal
2. View Daily Summary
3. Set Calorie Goal
4. Exit

Enter your choice: 1

Select Meal: 1.Breakfast 2.Lunch 3.Dinner 4.Snacks

Your choice: 2

Food Database:

1. Rice (1 cup) - 206 kcal | P:4g C:45g F:0.4g
2. Chicken (100g) - 239 kcal | P:27g C:0g F:14g

Select item (0 to stop): 2

--- Daily Summary ---

Total Calories : 860 / 2000 kcal

Protein : 54 g

Carbohydrates : 90 g

Fats : 28 g

Remaining : 1140 kcal

Note: The displayed values are examples. Actual values depend on the food items logged by the user.

8. Conclusion

The Calorie & Nutrition Tracker is a simple and effective console-based application developed using C programming. It demonstrates the use of menu-driven programming, structures, file handling, and input/output operations. This project helps beginners understand program control flow, data aggregation, and logical decision-making. It can be further enhanced by adding graphical progress bars, weekly reports, BMI calculation, exercise calorie burn tracking, and a larger food database with search functionality.